

Assignment 12

Submission by: **May 26th - 4pm**

1) [0.3 points]

Consider the following 3SAT formula:

$$\phi = (x \vee \bar{y} \vee \bar{z}) \wedge (\bar{x} \vee y \vee z) \wedge (\bar{x} \vee \bar{y} \vee \bar{z})$$

Construct the reduction to a 3CLIQUE instance

2) [0.4 points]

Show how to reduce $2SAT \leq_p CLIQUE$.

What can you conclude once you have a correct reduction?

3) [0.3 points]

Consider the following 2SAT formula:

$$\phi = (x \vee y) \wedge (\bar{x} \vee y) \wedge (\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee \bar{y})$$

Construct the reduction to CLIQUE instance.

Does the obtained graph contain a 4CLIQUE?

Does it contain a 3CLIQUE?

Argue why.

4) [0.5 point]

Let $DOUBLE - SAT = \{ \langle \phi \rangle \mid \phi \text{ has at least two satisfying assignments} \}$.Show that $DOUBLE - SAT$ is NP-Complete.

5) [0.3 points]

Consider the following 3SAT formula:

$$\phi = (x \vee \bar{y} \vee \bar{z}) \wedge (\bar{x} \vee y \vee z) \wedge (\bar{x} \vee \bar{y} \vee \bar{z})$$

Construct the reduction to a HAMPATH instance and highlight the Hamiltonian path.

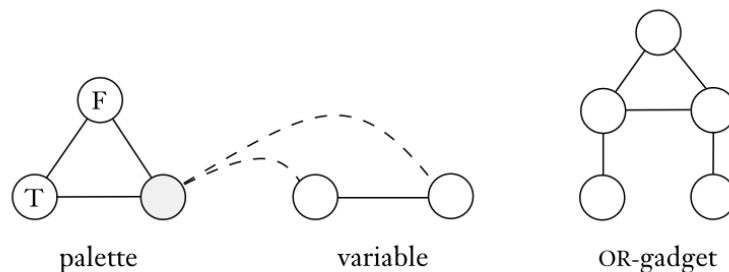
6) [1.5 point]

A *colouring* of a graph is an assignment of colours to the nodes such that no node is adjacent to another node with the same colour.

$$3COLOR = \{ \langle G \rangle \mid G \text{ is colourable with 3 colors} \}.$$

Show that 3COLOR is NP-Complete

Hint (proof sketch of construction): Reduce 3SAT to 3COLOR and use the following subgraphs. The graph needs to use three colours: True, False and Neutral (T,F,N). The palette will allow you to make sure that for each variable either x_i is coloured True and $\bar{x}_i$ is coloured False or viceversa. To achieve this you can connect the variables to the **common** palette. For each clause you need at least one variable to be True for the clause to be satisfied. The OR gadget will allow you to work this out.



For clause $(a \vee b \vee c)$ this OR-gadget will allow you to have the output node coloured True if either a , b , or c are coloured True. But if they are all coloured False (i.e., you don't have a truth assignment for the clause), then you will be forced to colour the output node False. Finally you can connect the output nodes to the palette in the right place.

